Hybrid representations (encoders with acoustic features and encoder combinations)

FINAL_V1 experiment group summary

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Results

- 32 of 32 conditions in this group are complete.
- Highest pooled AUPRC (descriptive, selection-biased among 32 conditions): H3 Logistic regression, AUPRC 1.000, AUROC 1.000.
- Lowest: H1 RBF SVM, AUPRC 0.901, AUROC 0.867.
- Range of AUROC over the group: 0.867 to 1.000; median 0.995.
- Speaker errors at threshold 0.5 (false negatives + false positives) per condition: minimum 1, median 1, maximum 10 (out of 47 speakers).
- Fold variability: median over conditions of the mean fold AUROC is 1.000 with a median SD of 0.000 over the five outer folds (9 to 10 speakers per fold).

Hybrid versus its constituent baselines (same model, same folds)

- 76 paired comparisons; the 95% bootstrap interval of the AUROC difference excludes zero in 5 (positive in 4, negative in 1); unadjusted and descriptive.
- Median AUROC difference (hybrid minus constituent): 0.000; range -0.122 to 0.102.

Reading the table

- Primary metrics are pooled speaker-level out-of-fold AUPRC and AUROC (47 speakers: 25 PD, 22 controls; prevalence 53.2% PD); F1, sensitivity, specificity and balanced accuracy use the fixed threshold 0.5.
- Intervals are percentile bootstrap intervals over speakers (2,000 resamples); with 47 speakers one person moves a metric by roughly 2 points. Comparisons are descriptive; no significance is claimed.
- Recording date and class are confounded in this corpus (every date holds one class; all 44.1 kHz audio is PD), so a high score does not show disease detection.

Table 1: Pooled out-of-fold speaker-level results

Representation	Model	AUPRC [95% CI]	AUROC [95% CI]	F1	Sens.	Spec.	Bal.acc.
H1	Logistic regression	0.998 [0.992, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
H1	Linear SVM	0.991 [0.967, 1.000]	0.987 [0.955, 1.000]	0.980	0.960	1.000	0.980
H1	RBF SVM	0.901 [0.796, 0.969]	0.867 [0.749, 0.954]	0.828	0.960	0.591	0.775
H1	XGBoost	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
H2	Logistic regression	0.997 [0.986, 1.000]	0.996 [0.984, 1.000]	0.980	0.960	1.000	0.980
H2	Linear SVM	0.989 [0.958, 1.000]	0.982 [0.935, 1.000]	0.980	0.960	1.000	0.980
H2	RBF SVM	0.956 [0.866, 1.000]	0.962 [0.891, 1.000]	0.893	1.000	0.727	0.864
H2	XGBoost	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
H3	Logistic regression	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980
H3	Linear SVM	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980
H3	RBF SVM	0.985 [0.946, 1.000]	0.973 [0.905, 1.000]	0.941	0.960	0.909	0.935
H3	XGBoost	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
H4	Logistic regression	0.996 [0.981, 1.000]	0.995 [0.977, 1.000]	0.980	0.960	1.000	0.980
H4	Linear SVM	0.992 [0.970, 1.000]	0.989 [0.960, 1.000]	0.980	0.960	1.000	0.980
H4	RBF SVM	0.986 [0.947, 1.000]	0.975 [0.912, 1.000]	0.960	0.960	0.955	0.957
H4	XGBoost	0.994 [0.978, 1.000]	0.993 [0.971, 1.000]	0.980	0.960	1.000	0.980
H5	Logistic regression	0.997 [0.985, 1.000]	0.996 [0.983, 1.000]	0.980	0.960	1.000	0.980
H5	Linear SVM	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
H5	RBF SVM	0.986 [0.947, 1.000]	0.975 [0.912, 1.000]	0.941	0.960	0.909	0.935
H5	XGBoost	0.993 [0.974, 1.000]	0.991 [0.964, 1.000]	0.980	0.960	1.000	0.980
H6	Logistic regression	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980
H6	Linear SVM	0.992 [0.969, 1.000]	0.989 [0.961, 1.000]	0.980	0.960	1.000	0.980
H6	RBF SVM	0.986 [0.947, 1.000]	0.975 [0.912, 1.000]	0.960	0.960	0.955	0.957
H6	XGBoost	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
H7	Logistic regression	0.997 [0.985, 1.000]	0.996 [0.983, 1.000]	0.980	0.960	1.000	0.980
H7	Linear SVM	0.997 [0.987, 1.000]	0.996 [0.984, 1.000]	0.980	0.960	1.000	0.980
H7	RBF SVM	0.984 [0.943, 1.000]	0.971 [0.901, 1.000]	0.941	0.960	0.909	0.935
H7	XGBoost	0.993 [0.974, 1.000]	0.991 [0.964, 1.000]	0.980	0.960	1.000	0.980
H8	Logistic regression	0.997 [0.987, 1.000]	0.996 [0.984, 1.000]	0.980	0.960	1.000	0.980
H8	Linear SVM	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980
H8	RBF SVM	0.986 [0.947, 1.000]	0.975 [0.912, 1.000]	0.941	0.960	0.909	0.935
H8	XGBoost	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.980	0.960	1.000	0.980